Star Kingdom Pipe Replacement

Complete model overview: what we built, what we left behind, and why

STAT 440 · Team PathFinder (Group 4) · Best public score **$77.67M** · 2 November 2026

At a glance

- **Best public score: $77.67M** = $250M budget − $139.54M construction − $32.79M leak costs.
- **The final model is not a machine-learning model.** It is a statistical survival model: an empirical leak rate by pipe age, adjusted for batch depletion, weighted by expected leak cost, and projected 4.5 years ahead.
- **We built, validated and submitted three gradient-boosting models.** Each predicted next year's leaks better than the age curve, and each produced a worse replacement plan, both in our backtests and on the leaderboard. We left them for that reason.
- **The largest late gain (about +$1.2M) came from understanding the data's structure:** leak bursts are laying cohorts reaching failure age. That led to a planning horizon long enough to get ahead of the 2030 burst.

1. The problem

The city must replace every iron pipe in its network, about 26,000 of 43,039 pipes, spending at most $10 million a year from a $250 million budget. The score is the budget minus construction cost minus the cost of leaks that happen before pipes are replaced. At $10 million a year, replacement takes about 14 years, and any pipe still waiting in line can leak.

A plan therefore has two jobs. It must **dig cheaply**, which depends on how pipes are grouped into circular work units, and it must **dig in the right order**, replacing pipes before they leak. Our inputs were the pipe inventory (material, surface, lay date, coordinates) and 14,113 leak records from 2019 to 2026.

2. The final model

The plan is built in four steps. The first three estimate how much each pipe is likely to cost us if it waits; the fourth turns those estimates into a dig order.

Step 1. Leak probability from age

From the 2019–2026 history we estimate, for each pipe age, the share of pipes at risk that leaked in a year: a life-table estimate of the leak hazard, the standard starting point in survival analysis. The curve has a sharp peak. Pipes aged 60–64 leak at about **13.8% a year**, against 1–4% at other ages. Each pipe's age is rolled forward through a **planning horizon of 4.5 years** (2027–2030 in full, 2031 at half weight), and the yearly chances are combined into one probability: 1 minus the chance of surviving every year in the window.

Step 2. Depletion adjustment

Iron pipes that touch end to end and were laid within 14 days of each other form a **batch**, almost certainly one construction job. Each pipe leaks only once and weak pipes fail first, so once most of a batch has leaked, its survivors are the sturdy ones. Each pipe's probability is multiplied by *(batch pipes not yet leaked + 1) / (batch size + 1)*: 1.0 for an untouched batch, about 0.36 for a batch of ten with seven already leaked.

Step 3. Leak cost

The expected repair cost of a leak, estimated from past leak costs by surface. Leaks under roads and structures cost far more than leaks under grass, and a handful of very expensive leaks dominate the total.

> **priority = P(leak within 4.5 years, from age) × depletion multiplier × expected leak cost**

Step 4. Circles and schedule

Nearby pipes are grouped into **13,814 merged circles** (clustering parameters 45 m and 50 m). A circle's value is the sum of its pipes' priorities. Circles are dug in order of **value per dollar of construction**, spending $10 million a year with unspent money carried forward. That order is the submission.

Final plan	Value	Comment
Circles dug	13,814	All iron pipes covered (checked before every submission)
Construction	$139.54M	$88M of it is the fixed $200/m charge for iron pipe, paid by every team
Leak costs	$32.79M	$13.71M of it in 2027, before much digging is possible
Public score	**$77.67M**	**Best of all submissions**

3. Why we left machine learning, and what we use instead

We did not start with a simple model. Most of the project's modelling effort went into gradient boosting, and the final model is simple because the evidence kept pointing that way.

3.1 What we built

Model	What it predicted	Public score
Gradient-boosting classifier, blended with the age curve	Will the pipe leak next year? (yes/no)	$73.31M
Gradient boosting on leak dollars (Poisson loss), blended with the age curve	Leak dollars next year	$75.92M
Pipe-year gradient-boosting hazard model (age, cohort, depletion, nearby leaks, location)	Leak probability each year, 2-year horizon	$68.24M
Final statistical model (age curve × depletion × cost)	**Leak probability over 4.5 years**	**$77.67M**

3.2 What the evidence showed

Every gradient-boosting model was **better at predicting next year's leaks** and **worse at building a plan**. Blending either model into the age curve raised the share of next-year leak dollars caught by the riskiest 10% of pipes, yet lowered the leak dollars a plan actually prevented. The pipe-year model showed the gap most sharply: on the leaderboard it beat the age model by $0.38M in year 1, then let year-2 leaks more than double ($8.11M against $3.14M), finishing $8.2M behind.

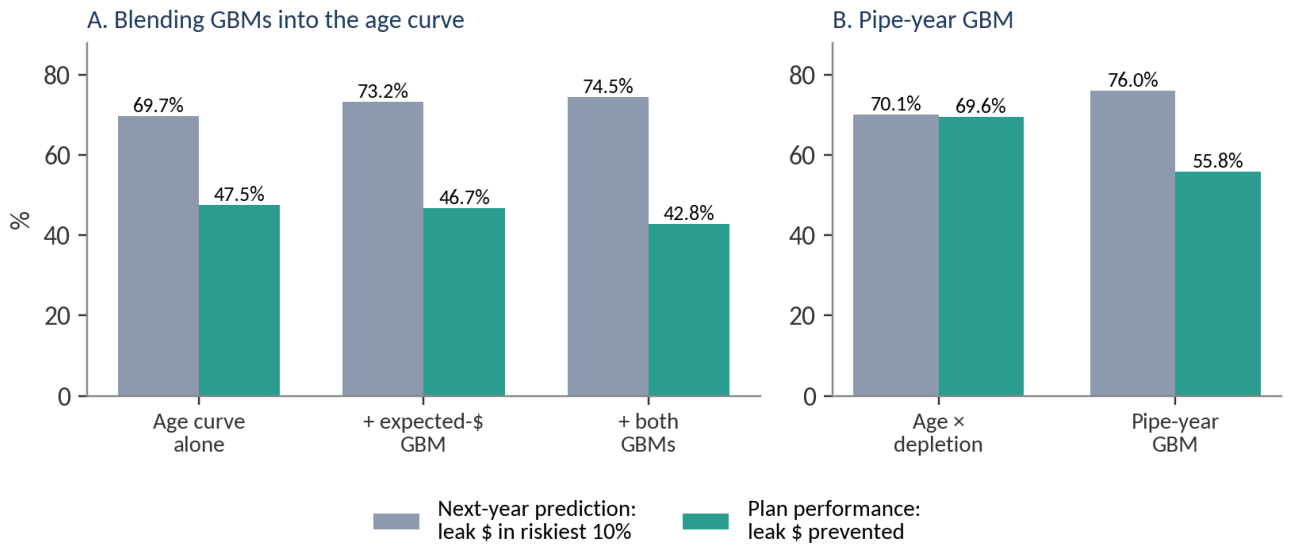


Figure 2. Better prediction, worse plans. Grey: share of next-year leak dollars caught by each model's riskiest 10% of pipes. Green: share of leak dollars a plan built from the model prevented in walk-forward backtests (mean of three windows). Panel A used our original backtest and panel B the corrected one (Section 4), so compare within a panel.

3.3 Why the simpler model wins here

1. **The plan needs the whole ordering, over several years.** The budget is spent over about five years on the pipes that matter, ranked by value per dollar across thousands of pipes. Getting next year's top 10% right is not enough; the order of everything behind them matters too.
2. **The machine-learning signals were short-lived.** Features such as recent nearby leaks say a lot about the next 12 months and little about years 2–4. The models spent the budget chasing imminent leakers and neglected the pipes that would fail the year after.
3. **The age curve projects forward naturally.** A pipe aged 58 today will be 61 in 2030, so its future risk can be read straight off the curve. That is what made the planning horizon possible (Section 5.5).
4. **The data is mostly an age-and-cohort process.** In the pipe-year model, shuffling age cost 0.109 AUC and lay year 0.072; no other feature cost more than 0.013. The extra features added little that age did not already carry.
5. **It is stable and explainable.** The final model has no fitted tuning parameters beyond the age curve and the cost table, so there was little to over-tune on the backtest windows, and every component can be checked on its own.

3.4 What machine learning still contributed

The work was not wasted. The validation pipeline that every later change went through was built during the gradient-boosting phase, the models' feature importances confirmed that age and cohort carry the signal, and their failure taught the project's main lesson: for a budgeted multi-year plan, prediction metrics and plan performance can disagree.

In one sentence: we built a full machine-learning pipeline with walk-forward validation, and for this budgeted multi-year planning problem a simpler survival model outperformed gradient boosting, because the plan needs a stable ordering over several years rather than sharp next-year predictions.

4. How every change was tested

- **Local scorer.** A replay of the grader's cost formula, so construction cost could be checked before submitting.
- **Walk-forward backtests on real leaks.** Pretend it is the start of 2023, 2024 or 2025, build a plan from earlier data only, and measure how much of the leak cost that actually followed it would have prevented.
- **A decision rule fixed in advance.** A change had to beat the current best in the most recent window *and* on average across the three windows.
- **A known noise level.** Plans that are equivalent in backtests swing by about ±$0.4M on the public leaderboard, so smaller differences were treated as ties.
- **A late correction.** The backtest originally gave no credit for digging a pipe in the same year it would leak, while the grader does. Fixing it changed the percentages but reversed no decision.

5. Everything we did to improve the score

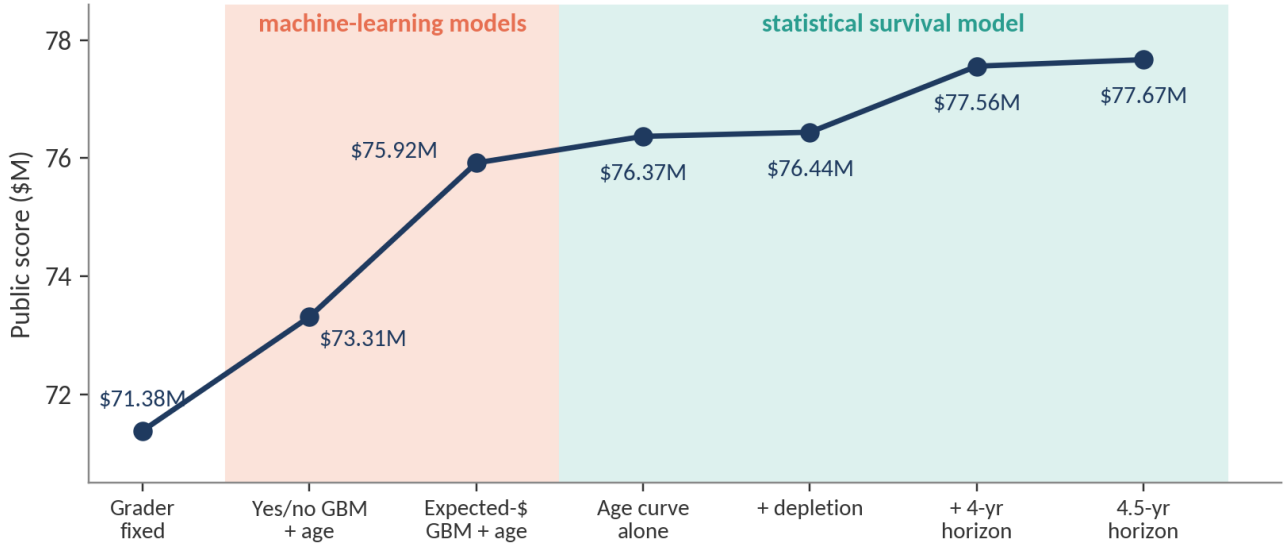


Figure 1. Public score after the grader was fixed. Earlier scores (−$4.12M and +$3.45M) were computed by a faulty grader and are not comparable.

5.1 Foundations (−$4.12M to $71.38M)

The local simulator was matched to the grader's rules: pipes that have already leaked need no digging, and circles with nothing left to replace cost nothing. Grouping pipes into shared circles instead of one circle per pipe cut construction from about $149M (the class baseline) to about $139M, the largest single saving in the project. A bug in the grader was then fixed and every team's score jumped, ours from +$3.45M to $71.38M.

5.2 Machine-learning phase ($73.31M to $75.92M)

A gradient-boosting classifier blended with the age curve reached $73.31M. Predicting leak *dollars* directly with a Poisson-loss model reached $75.92M; dollars matter because a few expensive leaks dominate. Several refinements were tested and rejected on backtest evidence: extra age and gap features, re-ranking every year, per-surface cost corrections, and bagging the expected-cost model.

5.3 Simplifying ($76.37M)

Backtests showed the age curve **alone** beat every blend with the gradient-boosting models, so they were dropped. A later blend probe ($76.11M) confirmed it.

5.4 Depletion ($76.44M)

We tested whether pipes fail by construction batch. Once more than half of a batch had leaked, its remaining pipes leaked at only **0.28, 0.35 and 0.27** times the rate their age predicted (relative to the network-wide level) in the 2023, 2024 and 2025 windows; batches with no leaks yet ran slightly above average. Our first idea, that each batch has its own failure age, found no support, and a version that raised risk for batches with many leaks pointed the wrong way. The depletion multiplier beat the age curve in every window of our original backtest (about +0.5 percentage points of leak dollars prevented on average) and stayed ahead on average under the corrected one.

5.5 Cohort waves and the planning horizon ($77.56M to $77.67M)

Leaks spiked in 2022 and 2026, and every grader breakdown showed a spike in 2030. A burst analysis showed these are **laying cohorts reaching failure age**, not network-wide bad years: the 1955–59 cohort was hit in 2022 and nearly silent by 2026, when the 1960–64 cohort took its turn, with 1965–69 already rising. That predicted the 1965–69 cohort would burst in **2030**.

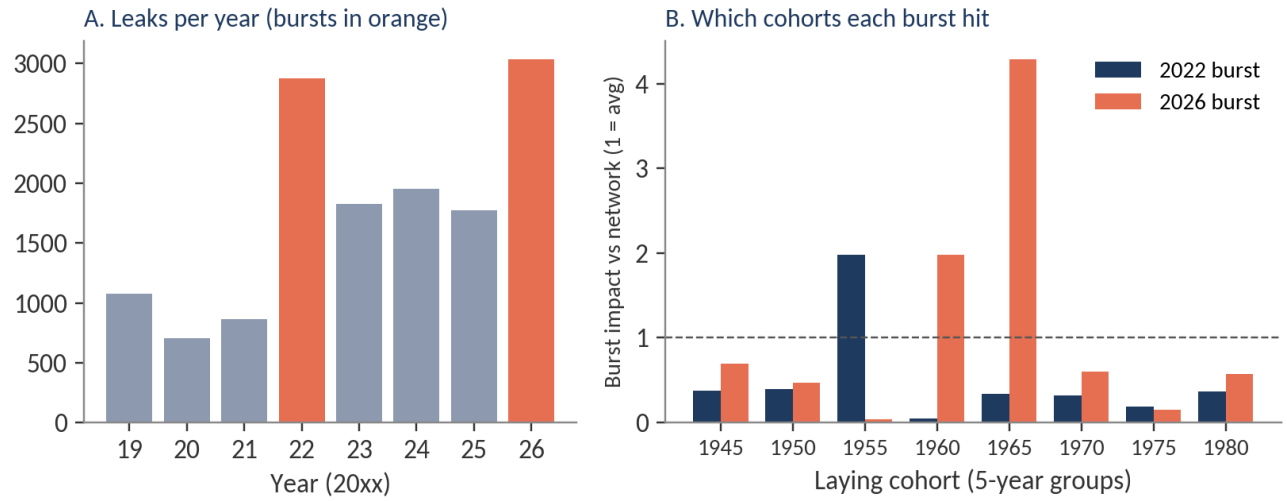


Figure 3. A. Leaks per year; 2022 and 2026 are bursts. B. How hard each burst hit each laying cohort relative to the network as a whole (1 = average). The burst moves from cohort to cohort.

The original plan looked only two years ahead, so a pipe aged 58 in 2027 looked as safe as one aged 56 even though it would reach failure age in 2029. Extending the horizon dug that cohort earlier. A sweep mapped the trade-off:

Horizon (years)	2	3.5	4	4.5	5	5.5
Construction ($M)	139.14	139.04	139.01	139.54	140.12	140.74
Leak costs ($M)	34.42	33.46	33.43	32.79	32.70	32.53
Public score ($M)	76.44	77.50	77.56	**77.67**	77.18	76.73

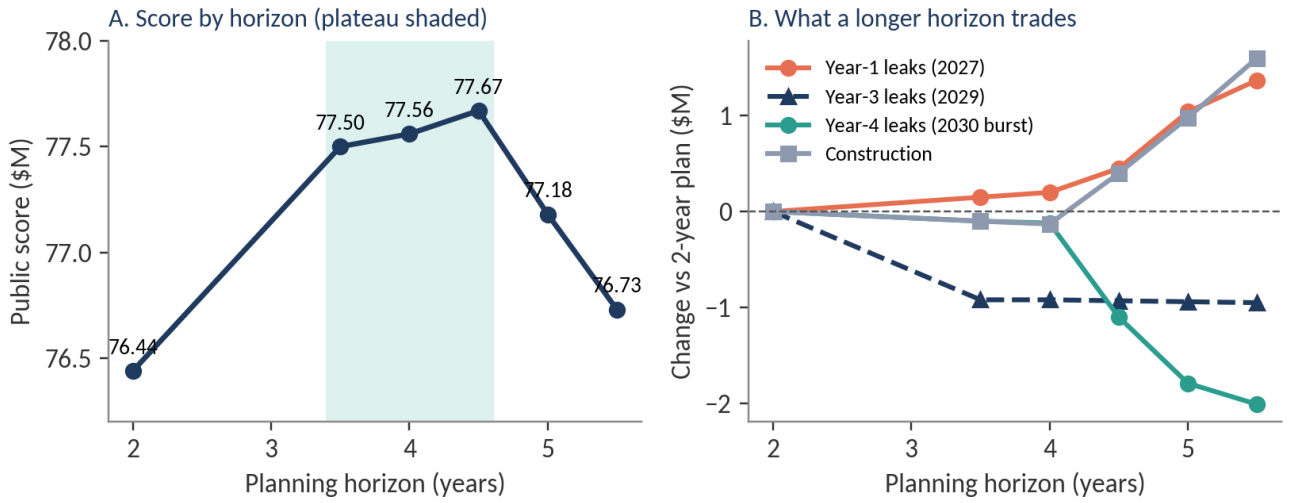


Figure 4. A. Score by horizon, with the 3.5–4.5 year plateau shaded. B. Changes against the 2-year plan: looking further ahead cuts 2029 and 2030 leaks, but year-1 leaks and construction rise.

Almost all of the gain arrives by 3.5 years, the first horizon that sees 2029, and the savings appear exactly in the years the cohort analysis predicted. Beyond 4.5 years the plan starts treating distant risk as urgent: year 1 gets worse, construction rises, and the score falls. The defensible conclusion is a **3.5–4.5 year plateau**; 4.5 years scored best, but its $0.11M lead over 4 years is within noise.

5.6 Construction (no further gains)

A cost breakdown of our scorer's pricing showed that most construction cost is fixed:

Construction component (2-year plan, our scorer)	Cost	Share
$200/m for iron pipe (paid by every team)	$88.14M	63.5%
Area term, (surface rate × area)$^{0.85}$	$48.90M	35.2%
$100 groundbreaking per circle	$1.38M	1.0%
$200/m for copper, brass and unknown pipe inside circles	$0.34M	0.2%

A circle optimizer working directly on the cost formula (shrinking, splitting and merging circles) saved only about 0.1%. A version tuned to a suspected alternative pricing rule scored *lower* on the leaderboard ($76.95M), which confirmed our scorer's pricing and showed the layout is at its floor.

5.7 Other ideas tested and rejected

- **Net-value ranking** (let cheap leaks happen to save digging): lower leak costs but higher construction, a wash at $77.44M.
- **Leak contagion** (a neighbour's leak raising your risk the same year): co-leak rates were near zero.
- **Bursts repeating in the same places:** the 2022 burst did not predict where 2026 hit; bursts follow cohorts, not locations.
- **A simulator to choose the horizon without the leaderboard:** it failed its own validation, missing the 2030 burst and overpredicting later years, so it was not used.

6. What each improvement contributed

Change	Effect	Type
Merged circles instead of one circle per pipe	About $10M less construction	Layout
Age-based ranking, simplified from gradient boosting	Core of the leak savings; +$0.45M over the best ML blend	Statistical model
Depletion multiplier	Small, consistent gain (+0.5 points in backtests)	Statistical model
Planning horizon of 3.5–4.5 years (cohort waves)	About +$1.2M, concentrated in 2029–2030	Statistical model

7. Lessons

1. **Simple, structural models beat complex ones here.** A smooth age curve that can be projected forward ordered pipes better over several years than any gradient-boosting model.
2. **Better prediction metrics did not mean better plans.** AUC and top-10% catch rates repeatedly disagreed with plan performance, because a budgeted multi-year plan needs the whole ordering right, not only next year's riskiest pipes.
3. **The biggest insights came from studying structure** (cohorts, depletion and the grader's year-by-year breakdowns) rather than from adding features.
4. **Discipline mattered.** Decision rules fixed in advance and a measured noise level kept us from chasing lucky public scores, and made the gains we kept believable.

8. Limitations

- **Unexplained construction differences.** The same circles cost between $139.0M and $143.5M in the grader depending on dig order, and we did not establish why.
- **Horizon chosen partly on public scores.** The 3.5–4.5 year plateau rests on a mechanism and on several submissions; the choice of 4.5 over 4 rests on a $0.11M difference, inside the noise, so private results may rank them differently.
- **The age curve underpredicts leak levels** by about a factor of two, and its projections beyond 2030 are unreliable because it does not account for how bursts use up each cohort's weak pipes.
- **The backtests cover at most four years,** because the data ends in 2026, so longer-range planning choices could not be validated directly.

Appendix A. Submission log (public leaderboard)

Submission	Public score	Outcome
Early plans (faulty grader)	−$4.12M, +$3.45M	Not comparable
After the grader fix	$71.38M	Starting point
Yes/no gradient boosting + age curve	$73.31M	Kept, later replaced
Expected-dollar gradient boosting + age curve	$75.92M	Kept, later replaced
Age curve alone (2-year horizon)	$76.37M	Replaced ML
Blend probe (a = 0.7, b = 0.2)	$76.11M	Rejected
Age curve × depletion (2-year horizon)	$76.44M	Kept
Pipe-year gradient-boosting hazard model	$68.24M	Rejected
Age × depletion, 4-year horizon	$77.56M	Kept
4-year horizon, net-value ranking	$77.44M	Wash
4-year horizon, re-optimized circles	$76.95M	Rejected
5-year horizon	$77.18M	Past the plateau
4.5-year horizon	**$77.67M**	**Best**
3.5-year horizon	$77.50M	On the plateau
5.5-year horizon	$76.73M	Past the plateau

Appendix B. Glossary

Term	Meaning in this project
Hazard	The chance that a pipe still in service leaks in a given year. Our age curve is an empirical hazard by pipe age.
Planning horizon	How far ahead the plan looks when judging a pipe's risk: P(leak at some point in the next H years).
Batch	Iron pipes that touch end to end and were laid within 14 days of each other; almost certainly one construction job.
Depletion	Each pipe leaks once and weak pipes fail first, so a group that has mostly failed has little risk left.
Cohort wave	A burst of leaks as one laying cohort (for example 1960–64) reaches its failure age.
Walk-forward backtest	Building a plan as of a past year with only the data available then, and scoring it on the leaks that followed.
O/E ratio	Observed leaks divided by the number the age curve expected; 1 means age explains everything.